TYPE CASTING

OTHER DATAYPE TO INT TYPE CONVERSION ---->¶

In [1]:
```
int(230.34)   #  int to float
```

Out[1]:
```
230
```

In [2]:
```
int(230.34, 110.23)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[2], line 1
----> 1 int(230.34, 110.23)

TypeError: 'float' object cannot be interpreted as an integer
```

In [3]:
```
int(true)  #int to bool
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[3], line 1
----> 1 int(true)

NameError: name 'true' is not defined
```

In [4]:
```
int(False)
```

Out[4]:
```
0
```

In [5]:
```
int(1 + 2j)   #int to complex
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[5], line 1
----> 1 int(1 + 2j)

TypeError: int() argument must be a string, a bytes-like object or a real number, no
t 'complex'
```

In [6]:
```
int('10')         #int to string
```

Out[6]:
```
10
```

In [7]:
```
int('ten')
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[7], line 1
----> 1 int(      )

ValueError: invalid literal for int() with base 10: 'ten'
```

OTHER DATAYPE TO FLOAT TYPE CONVERSION ---->

In [8]:
```
float(23)   #float to int
```

Out[8]: 23.0

In [9]:
```
float(True)    #float to bool
```

Out[9]: 1.0

In [10]:
```
float(1+2j)      #float to complex
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[10], line 1
----> 1 float(1+2j)      #float to complex

TypeError: float() argument must be a string or a real number, not 'complex'
```

In [11]:
```
float('20')    #float to string
```

Out[11]: 20.0

In [12]:
```
float("ten")
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
Cell In[12], line 1
----> 1 float(      )

ValueError: could not convert string to float: 'ten'
```

OTHER DATAYPE TO COMPLEX TYPE CONVERSION ---->

In [13]:
```
complex(10)      #complex to int
```

Out[13]: (10+0j)

In [14]:
```
complex(10, 20)
```

Out[14]: (10+20j)

In [15]:
```
complex(10, 20, 30 )
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[15], line 1
----> 1 complex(10, 20, 30 )

TypeError: complex() takes at most 2 arguments (3 given)
```

In [16]:
```
complex()
```

Out[16]: 0j

In [17]:
```
p, q = 20, 20
```

In [18]:
```
print(p)
print(q)
```

```
20
20
```

In [19]:
```
print(id(p))
print(id(q))
```

```
140724550346248
140724550346248
```

In [20]:
```
complex(10.3)  #complex to float
```

Out[20]: (10.3+0j)

In [21]:
```
complex(10.3, 23)
```

Out[21]: (10.3+23j)

In [22]:
```
complex(True)    #complex to bool
```

Out[22]: (1+0j)

In [23]:
```
complex(False, False)
```

Out[23]: 0j

In [24]:
```
complex('10')      #complex to string
```

Out[24]: (10+0j)

In [25]:
```
complex('20', '10')
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[25], line 1
----> 1 complex(    ,     )

TypeError: complex() can't take second arg if first is a string
```

OTHER DATAYPE TO BOOLEAN TYPE CONVERSION ---->

In [26]:
```
bool(15)     #bool to int
```

Out[26]:
```
True
```

In [27]:
```
bool(45.7)      #bool to float
```

Out[27]:
```
True
```

In [28]:
```
bool(3+6j)     #bool to complex
```

Out[28]:
```
True
```

In [29]:
```
bool("khan")      # bool to string
```

Out[29]:
```
True
```

OTHER DATAYPE TO STRING TYPE CONVERSION ---->

In [30]:
```
str(45)   #string to int
```

Out[30]:
```
'45'
```

In [31]:
```
str(34.6)     #string to float
```

Out[31]:
```
'34.6'
```

In [32]:
```
str(False)     #string to bool
```

Out[32]:
```
'False'
```

In [33]:
```
str(3+5j)     #string to complex
```

Out[33]:
```
'(3+5j)'
```

In [ ]:

PRINT() TYPE FUNCTIONS

(1) print() with string

In [34]:
```
num1=5
num2=10
num3=num1+num2
print ("The sum of",num1,'+',num2,'is=',num3)
```

```
The sum of 5 + 10 is= 15
```

In [35]:
```
name='Alfiya Zuber Khan'
proffession='Data Scientist'
experience='2 years'
#profile=name+proffession+experience
print ("Hi My name is",name, 'and my profile is',proffession,'and having experience
```

```
Hi My name is Alfiya Zuber Khan and my profile is Data Scientist and having experien
ce of 2 years
```

In [36]:
```
num1,num2,num3=10,20,30
print (num1)
print (num2)
print (num3)
```

```
10
20
30
```

In [37]:
```
num1=num2=num3=10,20,30
print (num1)
#print (num2)
#print (num3)
```

```
(10, 20, 30)
```

(2) print() with format method

In [38]:
```
num1=20
num2=30
num3=num1+num2
print ('the addtion of {} and {} is ={}'.format(num1,num2,num3))
```

```
the addtion of 20 and 30 is =50
```

(3) print() with End statement

It joins the line from end of one statement to the starting of another statements

In [39]:
```
print ('hello')
print ('world')
```

```
hello
world
```

In [40]:
```
print ('hello', end =' ')
print ('world')
```

```
hello world
```

(4) print with seprator

Here we use only print statements and under one print(), will pass multipple value and can separte these multi values with any thingd

In [41]:
```
print ('Hii','my name is','Alfiya','living in','Mumbai', sep= '-->')
```

```
Hii-->my name is-->Alfiya-->living in-->Mumbai
```

In [42]:
```
print ('Hii','my name is','Alfiya','living in','Mumbai', sep= '@')
```

```
Hii@my name is@Alfiya@living in@Mumbai
```

In [43]:
```
print ('Hii','my name is','Alfiya','living in','Mumbai', sep= '&')
```

```
Hii&my name is&Alfiya&living in&Mumbai
```

In [44]:
```
print(1,2, end = ' ')
print (3, '.',  sep='')
```

```
1 2 3.
```

In [45]:
```
print(1,2, end = ' ')
print (3, 'hree',  sep='')
```

```
1 2 3hree
```

In [ ]:

In [ ]:

In [ ]: